

MALIHA MEHNAZ

Innovation Team Manager

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PROFESSIONAL SUMMARY

AI Solutions Architect & Senior Python Engineer with **7+ years of experience** in **Python** development, **AI/ML engineering**, and **Generative AI** on **Azure cloud**. Expertise in **Python**, **FastAPI**, **Flask**, **Django**, **NumPy**, **Pandas**, **PyTorch**, **Scikit-learn**, **DevOps**, **MLOps**, **LLMOps**, **Docker** and **CI/CD pipelines**. Specialized in **LLM-based Python applications**, **RAG pipelines optimization**, **Agentic AI**, **vector databases**, and **Azure OpenAI**, delivering scalable enterprise AI solutions and cloud cost optimization.

PROFESSIONAL EXPERIENCE

Innovation Team Manager

2024/08 – 2025/12

Dexian Bangladesh 

- Architected and deployed 10+ enterprise-grade AI/ML solutions using SemanticKernel, LangGraph, LangChain, and Azure OpenAI improving operational efficiency by 12% & reducing manual workloads.
- Built standardized MLOps pipelines using Docker, MLflow, and CI/CD, reducing release cycles from 2 weeks to 5 days while improving deployment reliability.
- Developed retrieval-augmented generation (RAG) architecture using vector databases (FAISS, ChromaDB, Pinecone), enhancing model precision and increasing client adoption rate
- Led a cross-functional team of 8 AI engineers and data scientists, driving best practices in experimentation, deployment, and architecture design.

Senior Executive - Artificial Intelligence

2021/12 – 2024/07

Nitol Motors Limited 

- Built predictive models using Scikit-learn and XGBoost, saving ~\$9,000 annually.
- Improved customer segmentation accuracy by 28% through multithreading & multiprocessing, advanced feature engineering and model optimization.
- Developed Power BI dashboards for real-time analytics, cutting decision latency by 35%.
- Mentored a 5-member AI team, improving code quality, sprint velocity, and ML best-practice adoption by 20%.

Assistant Programmer

2018/12 – 2021/12

Golden Harvest Infotech Limited 

- Developed Python-based (OOP, Async,) automation scripts and APIs that saved over 300 staff hours monthly through workflow optimization.
- Created ML-powered analytics dashboards with Scikit-learn and Plotly, improving reporting efficiency by 18%.
- Preprocessed and cleaned 500K+ structured and unstructured data points.
- Optimized backend logic, reducing server response time by 20% and improving API performance.

CORE COMPETENCIES

Programming & Frameworks

Python, FastAPI, React, TensorFlow, PyTorch, Scikit-learn, LangChain, LangGraph, Vibe coding (VO, Firebase, Cursor, Github Copilot), Pydantic, SciPy

AI & Machine Learning

LLMs, Agentic AI Systems, RAG Pipelines, NLP, Deep Learning, Feature Engineering

Cloud & DevOps

Azure, AWS, Docker, CI/CD, MLflow, GitHub Actions, MLOps

Data & Visualization

SQL, PostgreSQL, MongoDB, Vector & Graph Databases, ETL Pipelines, Pandas, NumPy, Power BI, Plotly, Matplotlib, Seaborn, Grafana, Faiss, Data Warehousing, DataBricks, Redis

Leadership

Agile/Scrum, Solution Architecture, Team Mentorship, Stakeholder Management

SELECTED PROJECTS

ChatDocs – Document assistant enabling semantic Q&A.

Tech: React, FastAPI, PostgreSQL, LangChain, LLM, RAG

RoleplayAI – Real-time conversational simulator for training and evaluation.

Tech: Azure Whisper, Azure Cognitive Services, Snowflake, PyDub, LLM, Streamlit

Article Agent – Multi-agent assistant for automated generation, summarization and validation.

Tech: Python, LangGraph, Pinecone, DuckDuckGo Search

ATS Compatibility – Resume scoring using hybrid NLP and semantic search.

Tech: FastAPI, Gemini, Hybrid Search, JWT Authentication

AgentFlow – Natural language to pipeline automation tool for MLOps deployment.

Tech: LangGraph, Azure AI Foundry, Docker, GitHub Actions, Unit Testing (pytest)

EDUCATION

Master of Information Technology

2019 – 2021

University of Dhaka [🔗](#)

Bachelor of Science in Computer Science & Engineering

2013 – 2018

BRAC University [🔗](#)

KEY ACHIEVEMENTS

Awarded for **technical excellence & leadership** in driving digital transformation at **Dexian**.

Received **Performance Bonus** for accelerating innovation at **Golden Harvest InfoTech**.

LANGUAGE PROFICIENCY

IELTS General

Overall Band 7.5 [Listening: 8.0 | Reading: 7.5 | Writing: 7.0 | Speaking: 7.5]

PTE Academic

Overall Score 82 [Listening: 79 | Reading: 90 | Writing: 89 | Speaking: 76]